MCO-1D

Operator Manual

How to fly, talk to, and work with your ornithopter drone.

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Revision	22 — August 2026
New in v22	High-speed performance · Climb performance · Aerobatic gesture control · Wing coordination
Audience	Pilot / Operator — no engineering background required
Companion	Commissioning Manual v20 · Wing Addendum R21

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1 What This Manual Is For

This manual tells you — the pilot — everything you need to operate the MCO-1D ornithopter drone. It assumes the aircraft has been built and commissioned. No engineering knowledge required. You fly it like any aircraft.

◆ **NOTE** Every layer of software — wing configuration, learning, aerobatic routing, clap-and-fling — is completely invisible. You move the controls. The aircraft responds. That is the entire interface.

Aircraft at a Glance

Aircraft	MCO-1D metachronal cascade ornithopter SAR drone
Wings	8 — four tandem pairs, independently controlled
Weight	~4.1–5.1 kg flying
Stall speed	4.85 m/s (~9.4 kt)
Maximum speed	28 m/s (~54 kt)
Flapping	Up to 3.0 Hz — hard limit
Interface	Your controls → aircraft responds. No mode switches.

2 Equipment You Will Need

Aircraft side: MCO-1D commissioned, charged batteries, clear flying area, landing pad.

Ground station side:

- Laptop — any modern browser
- Python 3.9+ with `pip install aiohttp anthropic`
- Anthropic API key from console.anthropic.com — set as environment variable
- Files: mco_ws_bridge.py · mco_api_proxy.py · mco_ground_station_launcher.sh
- Interface: mco_ai_ground_station.jsx in Claude Artifacts

■ **CAUTION** Your API key is personal. Keep it in an environment variable only.

3 Starting the Ground Station

Three components must run before opening the interface. The launcher handles two automatically.

1. **Power on aircraft**
   Connect battery. Wait for solid green light (~10–15s).
2. **Connect to network**
   Join the aircraft Wi-Fi. Confirm with: `ping 192.168.7.1`
3. **Set API key**
   `export ANTHROPIC_API_KEY=sk-ant-...` (in the terminal you will use for Step 4)
4. **Run launcher**
   `bash mco_ground_station_launcher.sh 192.168.7.1` — leave running while you fly
5. **Open interface**
   claude.ai → attach mco_ai_ground_station.jsx → ask Claude to render it
6. **Check connection badge**
   Green DISARMED = ready. Red OFFLINE = check launcher and aircraft.

■ **CAUTION** If OFFLINE after 30s: check IP address, aircraft light, restart launcher.

4 The Conversation Interface

Type plain English. The AI understands intent, operates the aircraft, and explains results. You never need to know a command name.

Six steam gauges (top)	Airspeed, attitude, altitude, vertical speed, power, battery. Amber = normal. Red = attention needed
Status strip	Armed/safe, fault count, state, ALSS progress.
Conversation	Your messages right, AI replies left. Red alerts when needed.
Input box	Type and press Enter.

◆ **NOTE** The AI watches the instruments continuously. It speaks up about faults, low battery, and any condition needing attention — without you having to ask.

5 Pre-Flight Checks

"Ready to fly"
→ AI checks faults, battery (>21.5V), attitude hold, hover mode, airspeed sensor, GPS lock, wing guard calibration.

■ **CAUTION** Wing guard requires two measurements confirmed before first flight: fuselage width and TPU section start position. The AI will tell you exactly what to measure if not yet done.

Arming

"Arm it" → AI confirms once → arms → badge turns amber. Stand clear of all eight wings once armed.

6 Normal Flight Operations

Two modes: hover (below ~3 m/s) and fixed-wing cruise (above ~5 m/s). Transition is automatic.

HOVER	Lever = altitude. Stick = horizontal velocity. Release stick = hold GPS position.
FIXED-WING	Lever = thrust/braking. Stick = bank, pitch, yaw. Smooth blend between modes at 3–5 m/s.

◆ **NOTE** Say "take off", "land", "hold position", "come back" at any time. The AI handles everything else.

7 The Lever — Complete Guide

Forward = more power. Back = less power and braking. 50% = altitude hold.

ZONE	LEVER	WHAT HAPPENS
THRUST	0–5%	Maximum burst / sprint.
SPRINT	5–28%	High-speed sprint.
POWERED CRUISE	28–45%	Normal cruise.
ALTITUDE HOLD	45–55%	Holds altitude. 50% = exact hold.
GLIDE / RANGE	55–75%	Efficient glide. Kestrel here in good wind.
LIFT DUMP PARTIAL	75–80%	Rear wings reduce lift — steeper approach.
LIFT DUMP FULL	80–87%	Rear wings stop — full spoiler equivalent.
SHUTTLE BRAKE ½	87–93%	Progressive aerodynamic braking.
J-HOOK	93–97%	Post-stall pitch-up to hover. Needs 20m+.
SHUTTLE BRAKE FULL	97–100%	Maximum drag. ~2g. Auto-exits when slow.

8 Braking and Deceleration

All braking modes engage from lever position — no separate switch.

Lift dump (75–87%)	Rear wings reduce or stop → steeper approach at same speed.
Shuttle brake (87–97%)	All wings pitch to maximum drag. 1.5–2g deceleration. Auto-exits when slow.
J-hook (93–97%, >20m)	Post-stall pitch-up: nose high → decelerate → hover. Completes in <3s. Fails safe to shuttle brake
Emergency stop	Cuts all motors. Aircraft falls. Only if crash is certain and people are in the flight path.

■ **IMPORTANT** Emergency stop is not a landing method. Say "land" or "abort" for all normal situations.

9 How the Aircraft Learns

The MCO-1D improves continuously while flying. Every flight adds to its knowledge. You do not need to do anything for this to work.

Discovery	Tests small wing configuration variations during cruise. Keeps better-scoring variants. Runs automatic
Archive	Stores the best-ever configuration for every airspeed × wind × intent combination encountered.
Strategy genome	Learns which wing mechanisms work best for each manoeuvre — rolls, climbs, braking, hover — and

"That worked well"

→ Records a positive note. Configuration gets credit.

"The hover feels wobbly"

→ AI adjusts focus to crosswind stability.

■ **CAUTION** Say "we're done" after landing to save all session learning. Never disconnect the battery without saving.

10 Mission Focus

Tell the aircraft what to get better at. The learning system concentrates on what you describe.

WHEN YOU SAY…	WHAT IT DOES
"Gusty / rough air"	Turbulence stability focus.
"Hold position better"	GPS hold accuracy.
"Stay up longer"	Endurance — minimise power.
"SAR / rescue"	SAR hover and confined-space ops.
"Stormy conditions"	Storm resilience — maximum dihedral.
"Fast / top speed"	Maximum speed pursuit.
"Get there quickly"	Efficient high-speed transit.
"Fastest climb"	Best rate of climb.
"Clear the trees"	Steepest possible climb angle.
"Tight turns / aerobatics"	Roll authority and manoeuvre precision.
"Enable clap and fling"	Hover lift enhancement — ~25% in hover.
"Back to normal"	General optimisation resumes.

11 Wing Coordination and Clap-and-Fling

The eight wings coordinate automatically. A real-time guard monitors every port-starboard wing pair every tick. You feel none of this — the aircraft simply stays within safe limits and uses the physics when it helps.

What the guard does

The spar roots are fixed to the fuselage walls — spar-to-spar collision is geometrically impossible. Only the TPU winglet tips can make contact, and membrane contact at the right phase IS the clap-and-fling aerodynamic mechanism. The guard ensures contact only happens in the TPU zone, at the right stroke phase.

Clap-and-fling

Wings meet at the top of the upstroke and fling apart as the downstroke begins. The fling creates a leading-edge vortex captured on the next downstroke. Net hover lift enhancement: 15–25%. Accessed through the ground station:

"Enable clap and fling"

→ AI opens the clap gate. Wings contact at stroke reversal. Hover lift improves. Guard still validates timing every tick.

"Anti-phase cruise"

→ 180° port-starboard phase offset → 115° amplitude → ~20% more stroke volume per cycle. More efficient at cruise.

■ **CAUTION** Two measurements required before first flight and before pre-flight check will pass: fuselage width and TPU section start position. The commissioning engineer enters these during TC-01.

12 High-Speed Performance

The MCO-1D operates in three distinct speed regimes, each with different physics. The aircraft selects the appropriate wing configuration automatically as airspeed changes. The pilot just flies.

◆ **NOTE** No mode switch is needed for any speed regime. Advance the throttle and the aircraft configures itself. The wing changes happen over 2–3 wingbeats — the pilot feels the aircraft accelerating, not a configuration change.

Three speed regimes

SPEED RANGE	PHYSICS	WHAT THE AIRCRAFT DOES
0–15 m/s Efficient cruise	Strouhal-optimal (St = 0.20–0.40)	Stroke frequency and amplitude tuned so the wing moves at the optimal fraction of flight speed for
15–22 m/s Sprint	Fixed-wing morphing	Rear wing pairs stop and generate fixed-wing lift for free. Front pairs run as pure thrust generators.
22–28 m/s Maximum speed	Drag minimisation	Wings sweep fully aft, AoA reduced to minimum. Minimum frontal area, minimum drag. Equivalent

Speed focus options

You can tell the aircraft what to prioritise. These are optional — the aircraft performs well at speed with no focus set.

"Let's see how fast it can go / full throttle / maximum speed"
→ Top speed focus. Pursues 28 m/s Vne. Maximum sweep, minimum drag.

"Get there quickly / fast transit / speed and efficiency"
→ Fast cruise focus. Best speed per watt. Efficient at 15-22 m/s.

"Sprint / fast"
→ Sprint focus. Combines all speed configurations with structural safety.

13 Climb Performance

◆ **NOTE** The aircraft selects its climb configuration silently from your throttle and pitch inputs — exactly as a conventional aircraft responds to the same inputs. No mode switch. No announcement. Just fly it.

Two distinct climb optima

Best rate and best angle are fundamentally different. A pilot trained on conventional aircraft already knows which to use and when. The aircraft reads the same signals the pilot uses and routes accordingly.

BEST RATE OF CLIMB	Fastest altitude gain per unit time. Battery-efficient climbing. Use when climbing to SAR altitude or g How to fly it: high lever + moderate aft pitch. Hold that stick position. The aircraft settles at Vy.
BEST ANGLE OF CLIMB	Steepest possible slope. Minimum ground distance to gain altitude. Use when clearing trees, obstac How to fly it: high lever + harder aft pitch. The aircraft slows and steepens automatically.
SUSTAINED CLIMB	Best rate of climb within motor thermal limits. Can be maintained for 2–5 minutes continuously. Use How to fly it: high lever + gentle aft pitch. Aircraft finds the frequency that keeps motors within therm
ZOOM CLIMB	Trade airspeed for altitude. Enter at high cruise speed (18-22 m/s), pull back sharply. The aircraft co How to fly it: arrive fast, hard aft pitch. Happens automatically from speed.

◆ **NOTE** You do not select these modes. The aircraft reads your lever position, pitch input, and resulting airspeed — and routes internally to the correct wing configuration. From your perspective it simply responds to the controls as a well-designed aircraft should.

Focus options for climb (optional)

"Fastest climb / best rate of climb / climb to altitude"

→ Best ROC focus. Aircraft prioritises configurations that produce maximum vertical speed at Vy.

"Clear the trees / steep departure / obstacle clearance"

→ Best angle focus. Aircraft prioritises maximum gradient at Vx.

14 Aerobatic Flight

◆ **NOTE** The aircraft reads your stick inputs and routes to the correct aerobatic configuration silently. There is no aerobatic mode to engage. Fly it as you would any aerobatic aircraft. The aircraft gets better at every manoeuvre you practice.

How aerobatic routing works

Every tick, the aircraft measures how fast you are moving the stick — the urgency of your input — as well as the combination of roll, pitch, yaw, and lever. From these signals it classifies what you are doing and configures all eight wings appropriately. The configuration change happens over 1–3 wingbeats. You feel the aircraft responding, not a mode change.

What the aircraft recognises

WHAT YOU DO	WHAT IT IS	HOW FAST
Full roll + aft pitch together, fast throw	Snap roll	Config in 0.15s. Chainable immediately.
Full roll, fast throw, no pitch	Aileron roll	Config in 0.30s.
Roll + opposing yaw pedal	Lag roll — pure rotation without translation	Config in 0.25s.
Roll + slight constant aft pitch	Barrel roll — corkscrew path	Config in 0.40s.
Roll + firm aft pitch + high lever	Chandelle — climbing 180° turn	Config in 0.50s.
Roll + firm aft pitch, any lever	Wingover — altitude-preserving heading reversal	Config in 0.40s.
Fast throw + full roll + aft + V > 12 m/s + lever >70%	Immelmann — half loop then roll upright	Config in 0.35s.
Roll established, then push forward	Split-S — roll inverted then half loop	Config in 0.35s.
Slow deliberate full roll deflection	Slow roll	Config in 0.40s.

What is handled elsewhere

Banked turn tightening	Always active at any speed. Hard roll input at cruise speed produces progressively tighter turns aut
Zoom climb / pull-up	Read from lever + pitch. Enter from cruise speed with high lever and hard aft pitch.
Stoop / dive	High forward pitch + low lever. The aircraft configures for a fast diving descent.
J-hook deceleration	Lever 93-97% above 20 metres. Post-stall perch manoeuvre.

How the aircraft improves

Each manoeuvre type has a dedicated quality metric. Snap rolls are scored on sharpness — how instantly the roll fires. Barrel rolls are scored on altitude constancy throughout. Chandelles are scored on altitude gained per knot of airspeed surrendered. After enough repetitions the aircraft learns the optimal wing configuration for each manoeuvre at each airspeed and wind condition — and applies it automatically next time you fly the same manoeuvre.

◆ **NOTE** You do not need to practise manoeuvres deliberately for the aircraft to learn them. Every time you fly a chandelle, that flight scores. Every time you chain snap rolls, each one scores. The aircraft just gets better at what you actually use it for.

15 Forest Flight and Obstacle Avoidance

Obstacle sensing must be active and the sensor ring must be physically installed. Ask your commissioning engineer whether this hardware is fitted before entering confined spaces.

■ **CAUTION** Obstacle avoidance is banked for first flight — activate after the aircraft is flying reliably.

"Going into the forest / enable obstacle avoidance"

→ Activates the six time-of-flight sensor ring. Avoidance corrections add to your stick commands — your authority is fully preserved.

"The gap is X metres wide"

→ AI selects the correct passage strategy: >0.55m: flapping passage (upstroke phase hold) 0.20-0.55m: sideways passage (body rolls) 0.12-0.20m: ballistic tuck (wings stop, body coasts through)

"Go through the gap"

→ Begins gap negotiation.

"Approach through the trees"

→ Accipiter flap-flap-glide canopy approach.

"Land here / set it down on the ledge"

→ Vertical descent with position hold.

"Get out / climb out"

→ Confined exit — minimum span vertical climb.

16 Autonomous Learning Sorties

The aircraft flies, learns, and repeats with minimal human involvement. You define the flying area, start the programme, and swap batteries.

Class A (first 5)	Orientation. Hover, gentle cruise, rolls, climbs.
Class B (6-20)	Exploration. Efficiency, turns, lift dump, climbs, speed.
Class C (21+)	Performance. Full roll, VIFF, braking, J-hook, snap rolls, chandelles, best angle climbs, zoom, anti-phase c

1. **Define area**
   "Set practice area to [lat] [lon], 100m radius, 50m high"
2. **Start**
   Say "start the practice flights"
3. **Swap batteries**
   Aircraft lands, says "swap battery and re-arm". Swap, arm, next sortie begins. Learning saves automatically.
4. **End**
   Say "stop the practice flights"

■ **CAUTION** Safety observer should be present, especially for first Class C sorties. "Abort" or "land now" ends any sortie immediately.

17 Ceiling Exploration

■ **IMPORTANT** Ceiling exploration may reach altitudes requiring regulatory authorisation. Obtain all necessary permissions before proceeding.

The aircraft probes its altitude ceiling very slowly, holding each new altitude for 60 seconds before climbing further. Two explicit confirmations required.

1. **Complete checklist**

   "Arm ceiling exploration" → AI shows checklist. Confirm: airspace authorisation, NOTAM, observer, wind <5 m/s, battery full, aircraft in stable cruise.

2. **Confirm within 60 seconds**

   "Confirm ceiling exploration" → probe begins. If 60 seconds pass without confirmation, the arm expires.

"Abort the ceiling test"
→ Descends immediately.

18 After Flight

1. **"We're done" or "land and save"**
   AI saves all session learning and disarms. This single phrase does everything needed.

2. **Wait for DISARMED badge**
   Status badge returns to green. Then approach.

3. **Disconnect battery**
   Only after AI confirms disarmed and saved.

4. **Close ground station**
   Ctrl+C in the terminal stops everything cleanly.

◆ **NOTE** Disconnect without saving: only that session's discoveries are lost. All previous flights remain intact.

19 Troubleshooting

Start with the AI — it can diagnose most faults from the aircraft's data directly.

Interface shows OFFLINE

Launcher not running or aircraft not reachable. Check: launcher running, aircraft light green, same network, ping 192.168.7.1.

Pre-flight fails on battery

Charge battery. Must be above 21.5V.

Pre-flight fails on GPS

Open area, away from buildings. Wait 60-90s.

Pre-flight: wing guard not calibrated

Fuselage width or TPU start position not yet entered. Ask the AI "what measurements are needed" — it will tell you exactly what to measure.

Aircraft drifts in hover

"It's drifting" — AI checks attitude hold and GPS.

Aerobatic input not responding as expected

The aircraft requires V > 8 m/s for aerobatic routing. Check airspeed. At low speed, roll commands produce hover-mode responses.

Climb feels sluggish

Say "fastest climb". AI sets the best-rate focus. If already at focus, the aircraft may need more sorties to learn the optimal config.

Speed seems limited

Say "let's see how fast it can go". Top speed improves with Class C ALSS sorties as the aircraft discovers better high-speed configurations.

Lost connection during flight

Aircraft returns to takeoff point autonomously. Do not approach until landed and powered down.

Appendix A Useful Phrases

Use whatever words feel natural. These are examples only.

Starting

"Ready to fly"
→ Pre-flight checks.

"Arm it"
→ Arms after one confirmation.

"Disarm / We're done"
→ Disarms and saves.

Flight

"Take off / Let's go"
→ Takeoff.

"Land / Bring it down"
→ Landing.

"Hold position"
→ GPS hold.

"Come back"
→ Return to launch.

"Steep approach"
→ Lift dump — confined space landing.

Speed

"Top speed / Full throttle / As fast as possible"
→ Top speed focus.

"Get there quickly / Fast transit"
→ Fast cruise focus.

"Sprint"
→ Sprint focus.

Climb

"Fastest climb / Climb to altitude"
→ Best rate of climb focus.

"Clear the trees / Steep departure"
→ Best angle of climb focus.

"J-hook / Hard stop"
→ Post-stall deceleration to hover (>20m).

Aerobatics

"Tight turns / Aerobatics focus"
→ Aerobatics learning focus.

"That roll was good / Do that more"
→ Records positive note.

Wing coordination

"Enable clap and fling"
→ Opens clap gate. ~25% hover lift.

"Disable clap / Normal wings"
→ Closes clap gate.

"Anti-phase cruise"
→ 115° amplitude, more efficient cruise.

Forest flight

"Going into the forest"
→ Enables obstacle avoidance.

"The gap is X metres wide"
→ Sets gap width, selects strategy.

"Go through the gap"
→ Begins gap negotiation.

"Land here / on the ledge"
→ Perch landing.

"Get out / climb out"
→ Confined exit.

Learning

"How is the learning going?"
→ Progress summary.

"That felt wrong / sloppy"
→ Records concern, adjusts focus.

"Back to normal"
→ Clears focus.

ALSS

"Start practice flights"
→ Starts ALSS.

"How are sorties going?"
→ Progress report.

"Stop the practice"
→ Ends after current sortie.

"Abort / Land now"
→ Immediate landing.

Ceiling

"Arm ceiling exploration"
→ Step 1 — checklist + 60s arm.

"Confirm ceiling exploration"
→ Step 2 — begins probe.

"Abort ceiling test"
→ Descends immediately.

Ending

"We're done / End of day"
→ Saves learning, disarms.

"Save the learning"
→ Saves without disarming.

"Shut everything down"
→ Saves, disarms, close launcher.

Appendix B What the Aircraft Says

What the AI says on behalf of the aircraft.

"All checks pass. Ready to arm."
Pre-flight completed normally. Safe to arm.

"Arming now."
Stand clear of the wings.

"Alert: fault detected."
Ask "what is the fault?" immediately.

"Alert: battery low (X.Xv)."
Plan to land. Below 20V: land immediately.

"Wing guard not calibrated — fuselage width needed."
Measure spar-to-spar distance and report it. See commissioning manual.

"Clap gate open — dorsal mode."
Clap-and-fling active. Wings contact at stroke reversal. Guard validates every tick.

"Gap ahead: ~Xm wide — say "go through the gap" to enter."
Forward sensor detected an obstacle with clear sides. Gap passage available.

"Discovery is running."
Learning system actively testing configurations. Normal cruise behaviour.

"ALSS Sortie N complete. Swap battery and re-arm."
Autonomous sortie finished. Swap battery, arm, next sortie begins.

"Ceiling exploration — Step 1 of 2: Armed."
Checklist done. 60 seconds to confirm.

"Session saved. Safe to disconnect."
All learning saved. Safe to cut battery.

MCO-1D Operator Manual

Revision 22 — August 2026

New in v22: High-speed performance (§12), Climb performance (§13), Aerobatic flight (§14). All new capabilities are invisible to the pilot — the aircraft reads control inputs and routes internally with no pilot awareness.